

EGF as a non-invasive biomarker of endometriosis: a case control study

To the Editor, Endometriosis is a chronic inflammatory estrogen-dependent condition marked by growth of endometrial-like tissue outside the uterus that affects approximately 10% of women [1]. Advances in imaging have improved diagnosis of certain subphenotypes of endometriosis, but superficial peritoneal lesions with a revised American Society for Reproductive Medicine (rASRM) stage of I or II and patients with negative imaging findings or failure with empirical treatment can only be definitively diagnosed via surgical visualization [2]. While imaging can play an important role in diagnosis, imaging alone cannot rule out disease [3 , 4]. The lack of biomarkers of endometriosis leaves laparoscopy or laparotomy as the only option for objective diagnosis, contributing to the average 7-year delay from symptom onset to confirmed presence of endometriosis, leading to an average age of diagnosis in the fourth decade despite most patients experiencing symptoms before age 25 [5 - 7]. The identification of biomarkers to detect endometriosis could improve this timeline [8]. It would also provide insight into the pathophysiology of this disease, which could contribute to the development of personalized and more effective therapies. Urine is an ideal medium for non-invasive diagnostic and prognostic testing, rich with proteins and peptides that can serve as biomarkers. With a person's average urine output of approximately 1.5 L per day, urine collection is simple, economical and non-invasive [9]. We set out to discover potential urinary biomarkers that could differentiate between those with and without endometriosis. We used an unbiased and quantitative proteomics discovery approach, using urine samples from an existing cohort sampled from participants with surgically-confirmed endometriosis (Supplemental Table 1). iTRAQ was used in the discovery phase to identify and quantitate the urinary proteome of women with endometriosis and controls. We used an 8-plex iTRAQ protocol as previously described [9]. Employing a conservative 2.0-fold cutoff for the differential expression of urinary proteins from participants with endometriosis compared to samples from sex- and age-matched controls, we identified 26 proteins that were downregulated and 30 proteins that were upregulated in the urine of participants with endometriosis (Fig. 1). Total protein quantitation was determined using the Bradford Assay as previously reported [10 , 11]. Fig. 1 Four proteins identified via iTRAQ were selected for validation based on physiological relevance and availability of reliable ELISA kits. In participants 25-47 years old, normalized EGF levels (pg/ug) were found to be significantly lower in the urine of endometriosis participants compared to controls (control n = 24; endometriosis n = 22; P = 0.02, Supplemental Table 2). Predictive accuracy of urinary EGF for diagnosis of endometriosis in this group was 0.701 (Fig. 2 A). The other three analyzed proteins (RRBP1, MAFA and HPX) showed no ability to discriminate between participants with endometriosis and controls (data not shown) Fig. 2 (A) ROC curve for urinary EGF (pg/ug) in participants aged 25-47 with endometriosis compared to age-matched controls (p -value=0.01). (B) ROC curve for urinary EGF (pg/ug) in participants ages 25-47 with endometriosis and acyclic pelvic pain compared to age-matched controls without acyclic pelvic pain (P 0.001). (C) Quartiles of urinary EGF levels (pg/ug) with associated predicted probability of endometriosis in women 25-47 years of age and (D) in those same women who also have acyclic pelvic pain. Predicted probability was calculated for each quartile using Bayes formula assuming a 10% prevalence of endometriosis EGF ELISA results showed no significant difference in the concentration of EGF in urine between women with and without endometriosis when considering the whole cohort (Supplemental Table 2). A commonly reported symptom of endometriosis is acyclic pelvic pain [12]. Of those participants with endometriosis who reported the presence of acyclic pelvic pain, normalized urinary EGF levels were significantly lower than in the urine of controls who did not report the presence of acyclic pelvic pain (Supplemental Table 3). The accuracy of EGF in predicting the presence of endometriosis in women with acyclic pelvic pain is 0.767 (Fig. 2 B). Bayes formula was used with a population estimate of endometriosis prevalence of 10%, the prevalence of endometriosis in the general population, to determine PPV and NPV (Supplemental Table 4) [1]. A cutoff of 800 pg/ug resulted in a negative predictive value of 94.3%, sensitivity of 64% and specificity of 67% in the urine of endometriosis participants vs. controls aged 25-47 years. In those same samples from endometriosis participants who also experienced acyclic pelvic pain compared to control samples, the same cutoff of 800 pg/ug gave an even higher NPV of 96.8%, sensitivity of 80% and specificity of 68%. These NPVs suggest urinary EGF has promise as a clinical tool to rule out endometriosis in those with high levels, especially in the presence of acyclic pelvic pain. In addition to a binary cutoff value, we determined the probability of endometriosis based on quartile ranges of urinary EGF levels (Fig. 2 C and D). We found that women 25-47 years of age with low urinary EGF levels had approximately double the odds of endometriosis compared to those with higher urinary EGF levels (Fig. 2 C). Furthermore, when the presence of acyclic pelvic pain is considered, the risk increases approximately three-fold for those women with low urinary EGF levels (Fig. 2 D). Urine from participants with much higher urinary EGF levels (1200 pg/ug) had a 01% predicted probability of disease, further supporting the clinical potential of urinary EGF testing as a tool to rule out endometriosis. EGF shows promise as a potential novel, non-invasive biomarker for endometriosis that could enable early and frequent testing for this disease.

Supplementary Information Below is the link to the electronic supplementary material. Supplementary Material 1